

FACULTY OF BUSINESS AND IT

MBAI 5600G - Applied Integrative Analytics Capstone Project

Milestone 5 Report

**AI-Based Student Dropout Early Warning System for Higher Education:
Reproduction and Cross-Dataset Generalization on UCI and OULAD**

Student Group Name: 21

Team Members: Xinyu Wang, Wu Hanming

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1. Executive Summary

Milestone 5 is the optimization step after the Milestone 4 baseline models. The goal was not to add many unrelated algorithms, but to tune the model families that were already most relevant to the project. The work covers three modeling tasks: UCI multiclass reproduction, UCI binary early warning, and OULAD binary early warning. For UCI multiclass, we tuned XGBoost to stay close to the base-paper reproduction. For UCI early warning, we compared tuned Logistic Regression with tuned XGBoost. For OULAD, we tuned XGBoost because it was the strongest baseline model and the dataset includes behavioral and module-context features. All experiments used the same stratified train-validation-test design from Milestone 4. For the binary tasks, threshold tuning was also performed on the validation set instead of assuming that 0.50 was the best cutoff. The final test results showed that UCI early-warning Logistic Regression improved from 0.783 to 0.793 F1, and OULAD XGBoost improved from 0.729 to 0.738 F1. UCI multiclass XGBoost decreased slightly from 0.704 to 0.696 macro F1, so that result is treated as a caution rather than an improvement.

2. Novel Contribution Implementation

This milestone also addresses the concern that a single UCI/XGBoost/SHAP workflow would be too close to work that already exists. OULAD is now part of the primary analysis, not just backup data. The question is not only whether a model can identify students at risk of dropping out, but whether the useful signals stay similar when the data changes from a student-record dataset to a larger learning-analytics dataset.

The additional work was organized around three areas:

First, tuned models are compared against the Milestone 4 baselines rather than being reported by themselves. Second, the binary early-warning tasks include threshold tuning because a school would need to choose a cutoff before using a model for intervention. Third, SHAP output is grouped into shared feature families so UCI and OULAD can be compared even though their raw columns are different.

- UCI was used as the controlled reproduction and benchmark task.
- OULAD was used as the larger behavioral dataset containing early assessment and platform engagement variables.
- The shared feature-family comparison gives a practical way to compare explanations across datasets with different columns.

3. Advanced Model Development

The modeling workflow used the leakage-aware tables developed in Milestone 4. The project has three modeling tasks and four tuned model runs, and each task kept the same stratified 60/20/20 train-validation-test split. The first 60% of the data was used to fit candidate models during tuning. The validation 20% was used to choose hyperparameters and, for binary tasks, the classification threshold. The final 20% was left untouched until test evaluation. After the best candidate was selected, it was refit on the combined training and validation data before final test results were recorded. Preprocessing

stayed inside scikit-learn pipelines so the validation and test sets did not leak into training. Numeric missing values were filled with the training median, and numeric values were clipped at the 1st and 99th percentiles. Categorical missing values were filled with the most frequent training category and then one-hot encoded. Logistic Regression used `StandardScaler()`, while `XGBClassifier()` used the clipped numeric values without scaling.

Task	Rows	Type	Target	Features	Advanced model(s)
UCI multiclass reproduction	4,424	multiclass	Target	36 features	XGBoost
UCI binary early warning	4,424	binary	is_attrition	30 features	Logistic Regression, XGBoost
OULAD binary early warning	32,593	binary	is_attrition	51 features	XGBoost

The main modeling workflow is in `milestone5_model_development.py`. A shorter companion file, `modeling_results_walkthrough.py`, loads the saved results and walks through them in the same step-by-step style as the course example notebooks.

4. Hyperparameter Optimization Report

Hyperparameter tuning used focused grid searches rather than very large search spaces. This was a practical choice because the project still needed to run on a local laptop. For XGBoost, the search focused on tree count, tree depth, learning rate, minimum child weight, and class-weight adjustment for binary tasks. For Logistic Regression, the search focused on regularization strength while keeping balanced class weights.

Task	Model	Cand.	Metric	Val. score	Cutoff	Best settings
UCI multiclass reproduction	XGBoost	16	macro F1	0.727	N/A	trees=260; depth=5; lr=0.07
UCI binary early warning	Logistic Reg.	5	F1	0.776	0.50	C=0.05; balanced classes
UCI binary early warning	XGBoost	16	F1	0.787	0.46	trees=260; depth=3; lr=0.03
OULAD binary early warning	XGBoost	16	F1	0.744	0.60	trees=320; depth=5; lr=0.03

For the multiclass task, macro F1 was used because it measures performance across classes more evenly than accuracy. This matters because the Enrolled class is smaller. F1 was used for the two binary early-warning tasks because the project cares about both catching at-risk students and avoiding too many false alerts. Threshold values for the binary classifiers ranged from 0.20 to 0.80. The selected threshold prioritized F1, with recall and balanced accuracy used as tie-breakers when needed.

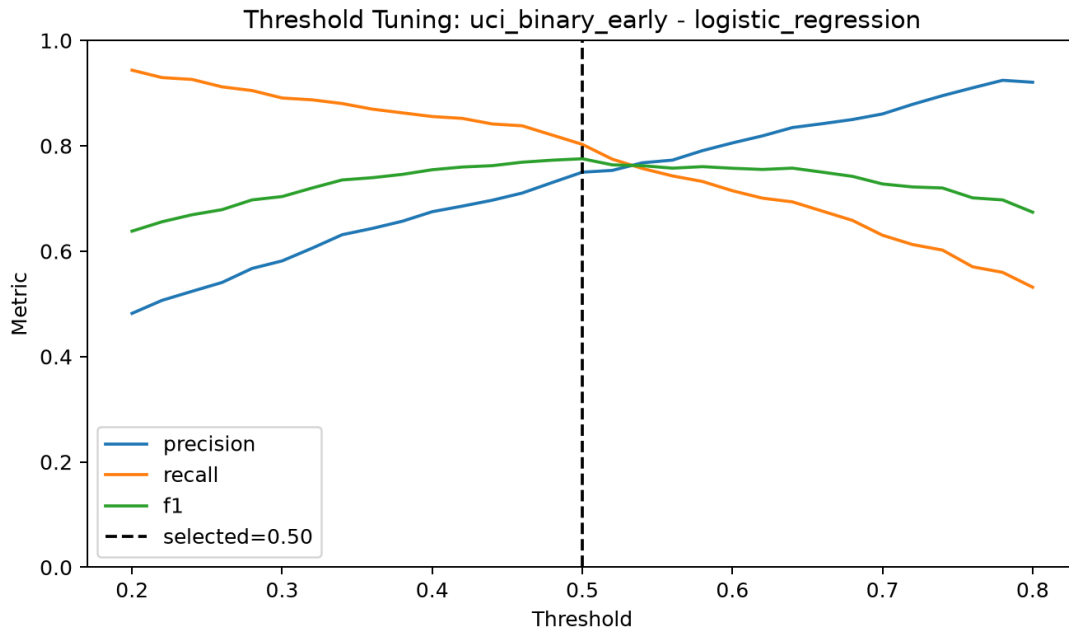


Figure 1. Validation threshold tuning for the UCI binary early-warning Logistic Regression model.

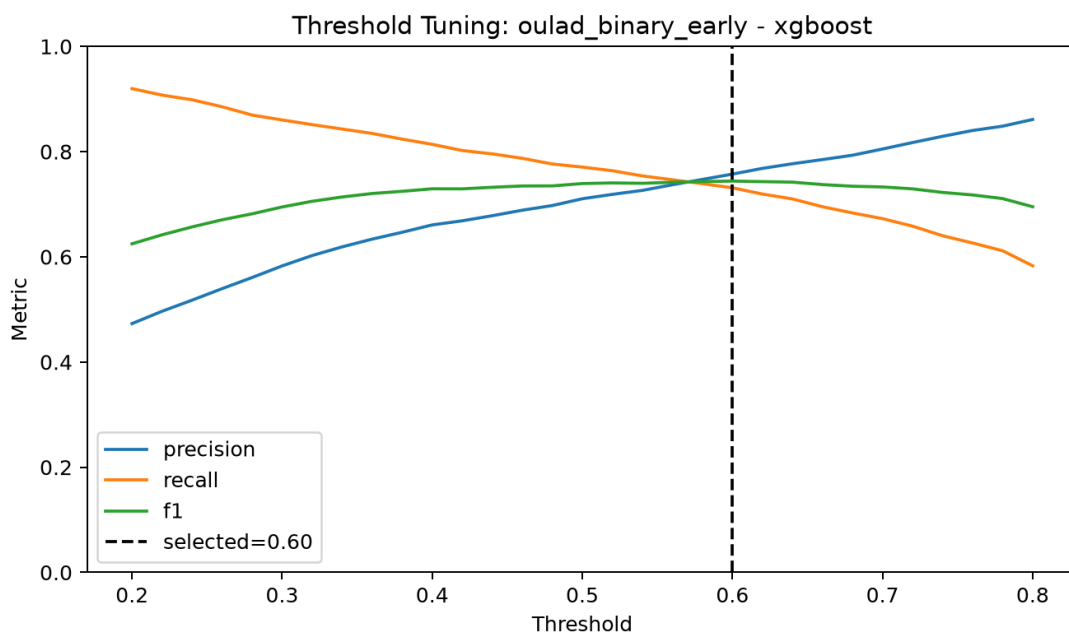


Figure 2. Validation threshold tuning for the OULAD binary early-warning XGBoost model.

5. Comparative Model Evaluation

Each tuned model was compared with its Milestone 4 baseline using the same held-out test split. This keeps the comparison fair. Validation scores were used to choose settings, but the test score is the main evidence for whether tuning actually helped.

Task	Tuned	Base	Eval.	Base	Tuned	Diff.	Acc.	ROC AUC
UCI multiclass	XGBoost	XGBoost	macro F1	0.704	0.696	-0.008	0.766	N/A
UCI early warning	Logistic Reg.	Logistic Reg.	F1	0.783	0.793	0.010	0.862	0.911
UCI early warning	XGBoost	Logistic Reg.	F1	0.783	0.767	-0.016	0.840	0.909
OULAD early warning	XGBoost	XGBoost	F1	0.729	0.738	0.009	0.841	0.885

The clearest improvement was the UCI binary early-warning Logistic Regression model. Its F1 increased from 0.783 to 0.793 without making the model harder to explain. OULAD XGBoost also improved, moving from 0.729 to 0.738 F1. This is useful because OULAD is larger and includes behavior data, so it is closer to the type of dataset an online early-warning system might use. The tuned UCI multiclass XGBoost did not beat its baseline on the held-out test data, so it is treated as a cautionary result rather than a successful optimization result.

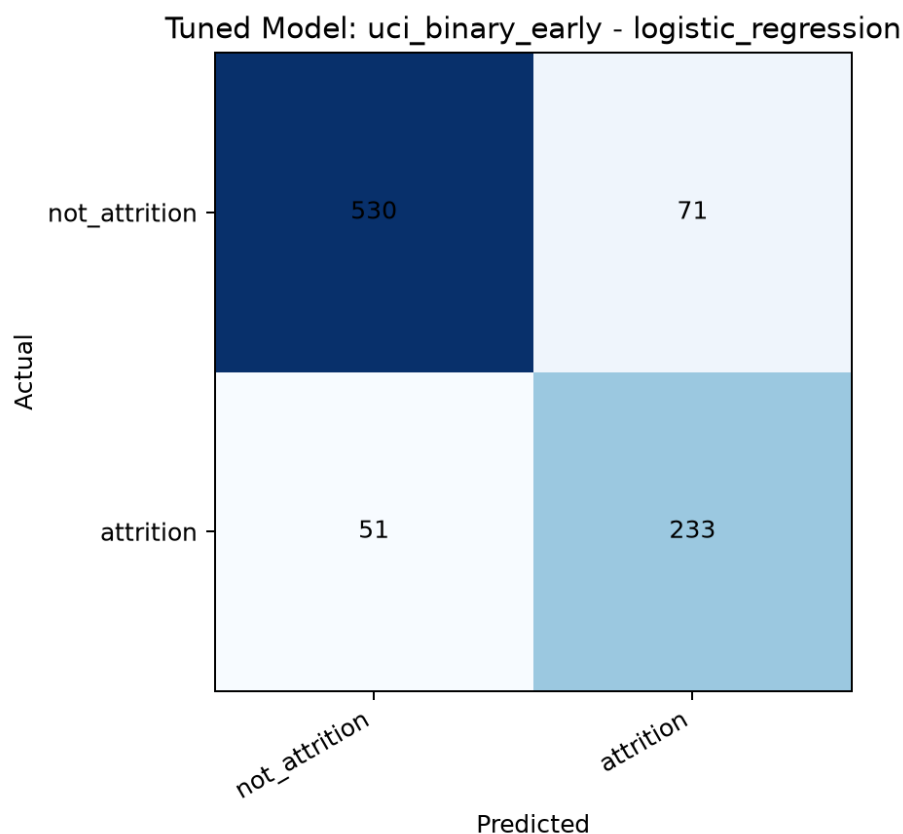


Figure 3. Tuned UCI binary early-warning Logistic Regression confusion matrix.

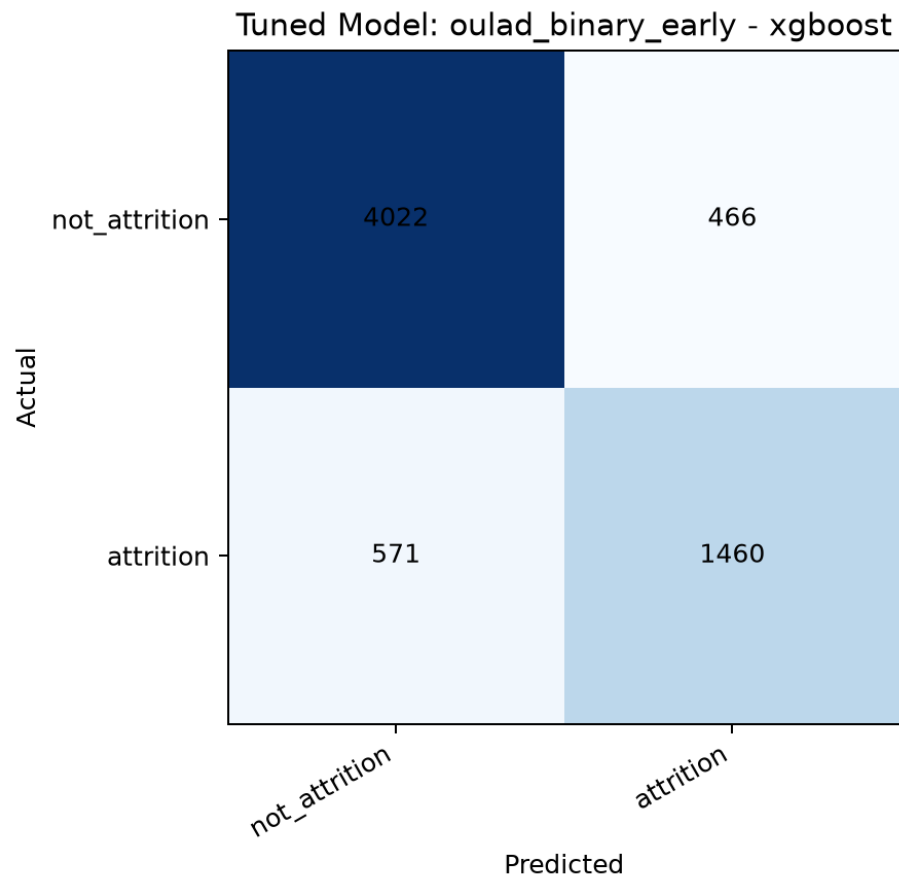


Figure 4. Tuned OULAD binary early-warning XGBoost confusion matrix.

6. Model Interpretation and Insights

For interpretation, we reviewed feature-importance outputs from the tuned models and used SHAP summaries for the XGBoost models.

SHAP helped compare feature-family patterns between UCI and OULAD, not just individual column names.

Model	Top SHAP feature	Mean absolute SHAP
UCI binary early XGBoost	Curricular units 1st sem (approved)	1.015
UCI binary early XGBoost	Tuition fees not up to date	0.340
UCI binary early XGBoost	Age at enrollment	0.272
UCI binary early XGBoost	Curricular units 1st sem (grade)	0.255
UCI binary early XGBoost	Curricular units 1st sem (enrolled)	0.199
OULAD binary early XGBoost	Early assessment submission ratio	0.841
OULAD binary early XGBoost	Highest early assessment score	0.194
OULAD binary early XGBoost	Early weighted-score ratio	0.192
OULAD binary early XGBoost	Early weighted-score sum	0.188
OULAD binary early XGBoost	Studied credits	0.131

For UCI, the explanation pattern was mostly what we expected from a student-record dataset. Early academic progress carried the strongest signal, especially approved curricular units and first-semester grade. Tuition status and age also appeared in the top features, which suggests that academic progress is not the only information used by the model.

Task/model	Feature family	Share of total SHAP
UCI early XGBoost	Early Academic Progress	43.8%
UCI early XGBoost	Financial Support	20.4%
UCI early XGBoost	Demographics	12.3%
UCI early XGBoost	Program Setup	10.7%
UCI early XGBoost	Other	4.5%
OULAD early XGBoost	Early Academic Progress	54.5%
OULAD early XGBoost	Early Engagement	19.1%

OULAD early XGBoost	Program Setup	17.4%
OULAD early XGBoost	Demographics	6.3%
OULAD early XGBoost	Prior Preparation	2.3%

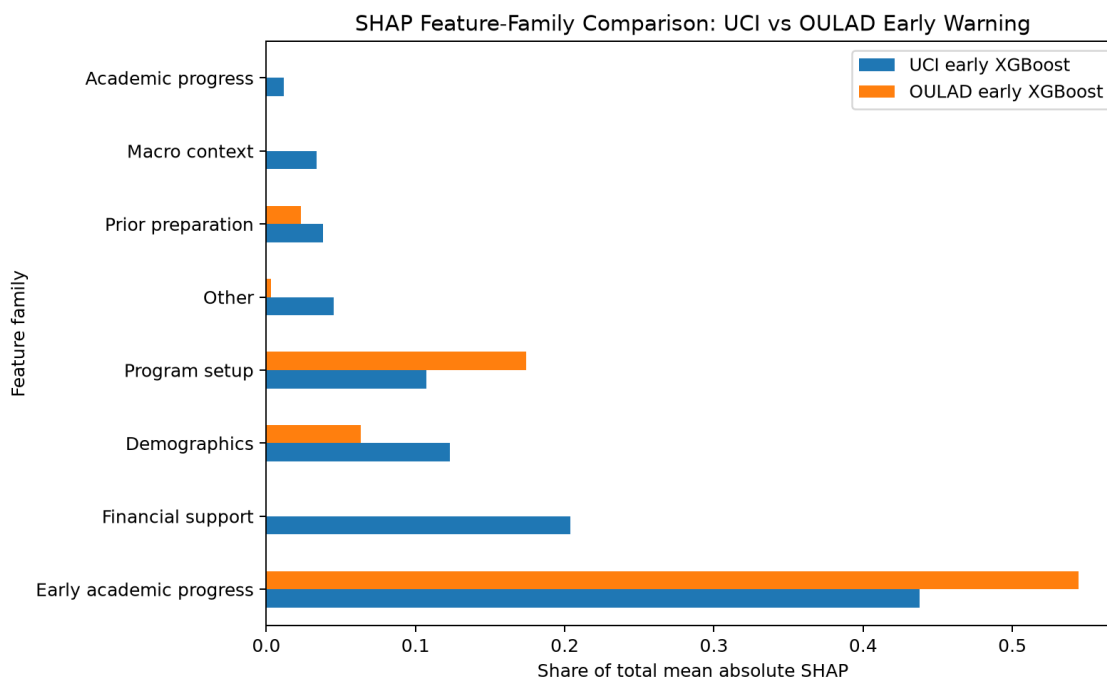


Figure 5. Feature-family SHAP comparison between UCI and OULAD early-warning XGBoost models.

The OULAD results pointed to a different kind of signal. Early assessment submission ratio, early assessment scores, weighted assessment progress, studied credits, and module context all appeared as important. These are the types of variables that could be monitored during a course in an online-learning setting.

7. Complexity vs. Performance Trade-off Assessment

The optimization results made the complexity trade-off clearer. XGBoost is more flexible than Logistic Regression, and that flexibility can help when the dataset contains many behavior and course-context variables. The cost is that it needs more tuning and is harder to summarize. In the UCI early-warning task, the simpler tuned Logistic Regression model was easier to defend because it had the best test F1 and remained easier to explain.

Context	Model	Performance trade-off	Decision
UCI binary early warning	Tuned Logistic Regression	Best test F1 in this task; simple and stable	Recommended candidate for UCI early-warning deployment
UCI binary early warning	Tuned XGBoost	Lower test F1 than baseline Logistic Regression despite high validation score	Useful for SHAP comparison, but not preferred for deployment
OULAD binary early warning	Tuned XGBoost	Improved test F1 and handles richer behavioral features	Recommended candidate for OULAD advanced modeling
UCI multiclass reproduction	Tuned XGBoost	Validation improved, but test macro F1 declined slightly	Reported as a robustness caution, not a final improvement

A safe conclusion for Milestone 5 is that the most complex model is not automatically the best model. Model choice depends on the dataset and the use case. Logistic Regression is the strongest current candidate for UCI early warning. XGBoost is still justified for OULAD because it improved performance and can handle interactions across assessment, engagement, and course-context features.

8. References (IEEE Style)

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